Air Pollution Detector Using Arduino UNO

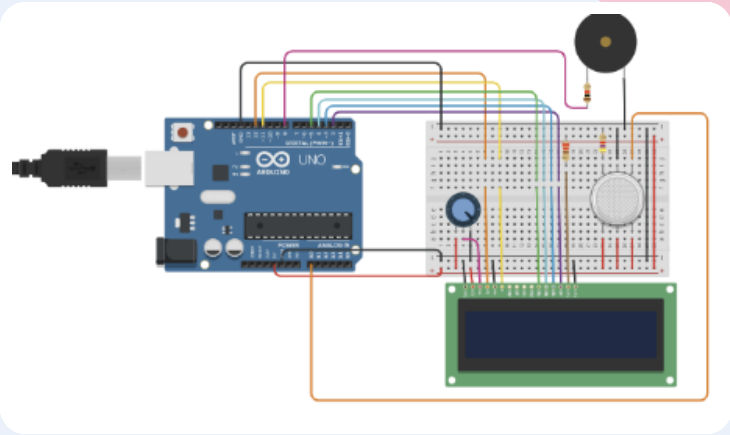
A Smart Prototype for Monitoring Air Quality

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Circuit diagram

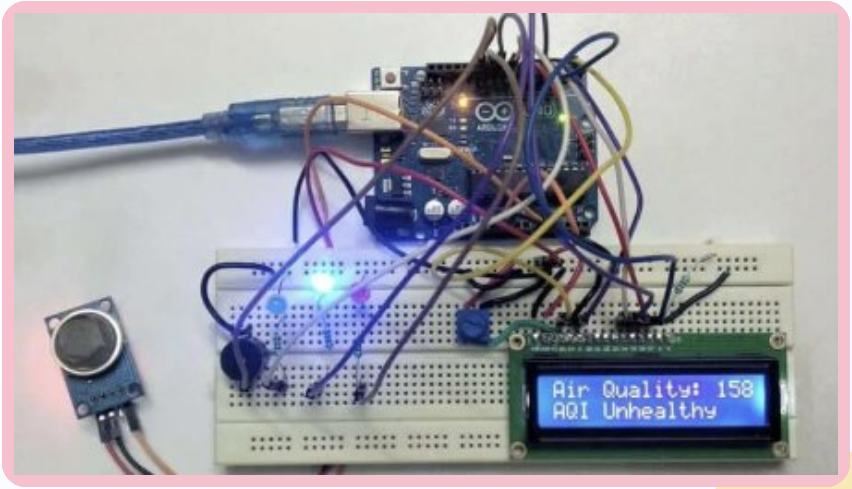
Introduction

Air pollution comes from vehicle emissions, industrial smoke, burning waste and harmful gases.

It affects human health and the environment.

This project is an Arduino UNO based prototype that senses gases and smoke in the air.

Readings are shown on an LCD screen.



Our prototype

Empathy: Who Is Affected?



Who is affected

Children and elderly people

People with asthma

Traffic police and outdoor workers

Families near industries



What they feel

Cough and breathing trouble

Eye and throat irritation

Headaches and tiredness

Worry about long-term health



What they need

A simple way to know when air is unhealthy

An affordable device

Clear and instant readings

Problem Statement

Vehicle smoke, industrial emissions and burning waste release harmful gases.

Many harmful gases cannot be seen or smelled.

People often notice the danger only after health effects begin.

Professional monitoring stations are costly and not available everywhere.

We need a low-cost way to monitor changes in air quality.

Why Arduino and Gas Sensors?

Low Cost

Arduino and MQ sensors are cheap and easy to find.

Easy to Build

Simple wiring on a breadboard, with easy programming.

Real-Time Readings

Sensor changes appear on the LCD within moments.

Expandable

New sensors, alerts and Wi-Fi can be added later.

Solution and Working

An Arduino UNO reads the signal from the MQ gas sensor placed in the air.

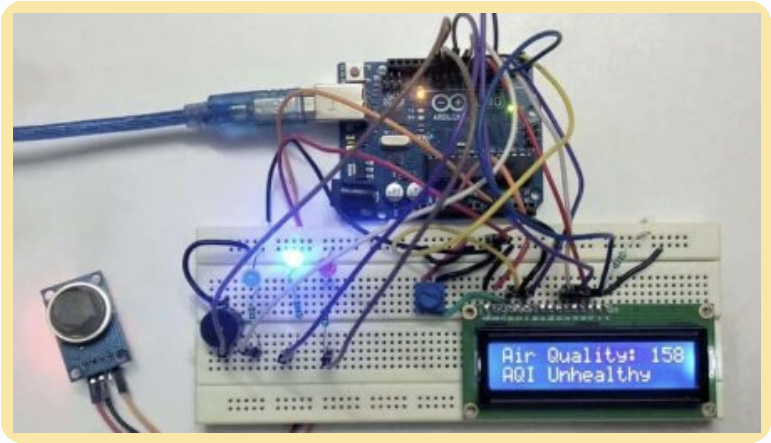
The Arduino processes the reading and shows it on the LCD.

A buzzer gives an alert when the reading is high.

Components: Arduino UNO, MQ gas sensor, LCD, buzzer, potentiometer, resistors, breadboard, jumper wires, USB cable.



Result



Working prototype with LCD output

The LCD shows the sensor reading and an air-quality status.

Values change when the sensor is exposed to smoke or gases.

Test Condition	Sensor Reading	LCD Output
Clean air (baseline)	To be measured	To be measured
Safe smoke source	To be measured	To be measured
Other safe test	To be measured	To be measured

Enter only actual measured values. Readings are not calibrated pollutant concentrations.

Advantages and Future Scope

Advantages

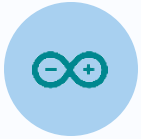
- Low-cost prototype
- Easy to assemble with common parts
- Real-time readings on the LCD
- Buzzer alert for high readings
- Good for learning and demos

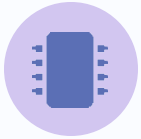
Future Scope

- Wi-Fi or Bluetooth connectivity
- Store readings and plot graphs
- Calibrate sensors for reliable estimates
- Add more sensors (dust, temperature)
- Mobile app alerts

Conclusion

The project shows air-quality monitoring using Arduino UNO.
The MQ gas sensor responds to gases and smoke.
Readings appear on the LCD, and the buzzer alerts on high values.
It is a useful learning platform for electronic monitoring systems.
Future upgrades can make it more informative and reliable.


Arduino


MQ Sensor


Clean Air


Environment


Health

Thank You!

Any Questions?